

Karnatak University
MSc/14/5762

THE FEDERAL UNIVERSITY OF TECHNOLOGY, P.M.B. 704, AKURE
DEPARTMENT OF MATHEMATICAL SCIENCES



Name of Examination: B.Tech in Industrial Mathematics
 Session: 2014/2015
 Course Code/Title/Unit: MTS 316/Engineering Mathematics II/ 3 Units
 Instruction: Answer Any Four (4) Questions
 Total Obtainable Score: 70 Marks
 Time Allowed: 2 Hours, Date: September, 2015

1. (a) Show that, if $-\pi \leq x \leq \pi$, and k is non-integer

$$\cos kx = \frac{2k}{\pi} \sin k\pi \left\{ \frac{1}{2k^2} + \sum_{r=1}^{\infty} (-1)^{r-1} \frac{\cos rx}{r^2 + k^2} \right\}$$

Hence, by putting $x = \pi$, and $k\pi = \theta$, show that

$$\cot \theta = \frac{1}{\theta} - \sum_{r=1}^{\infty} \frac{2\theta}{r^2\pi^2 + \theta^2}$$

- (b) Prove that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

2. (a) Show that $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
 (b) Evaluate the following (i) $\Gamma(\frac{3}{2})$ (ii) $\Gamma(-\frac{1}{2})$ (iii) $\int_0^{\infty} x^3 e^{-x} dx$
 (c) Prove that $B(m, n) = B(n, m)$
3. (a) Write the Hermite equation $y'' - 2xy' + 2xy = 0$ in Sturm-Liouville form.
 (b) Differentiate between ordinary and partial differential equations
 (c) When is a partial differential equation said to be (i) Elliptic (ii) Parabolic (iii) Hyperbolic? Give an example in each case
4. (a) Solve the vibrating string problem $\frac{\partial^2 u}{\partial t^2} = 3 \frac{\partial^2 u}{\partial x^2}$, $0 < x < \pi$, $t > 0$, $u(0, t) = u(\pi, t) = 0$, $t \geq 0$, $u(x, 0) = 3 \sin 2x + 12 \sin 13x$, $0 \leq x \leq \pi$, $\frac{\partial u}{\partial t}(x, 0) = 0$, $0 \leq x \leq \pi$
 (b) Verify that $u(x, t) = \sin x \cos kt$ is a solution of the wave equation $u_{xx} = \frac{1}{k^2} u_{tt}$
5. (a) Represent the function $f(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi \\ 0, & \text{otherwise} \end{cases}$ by Fourier integral formula and verify that
- $$\sin x = \frac{1}{\pi} \int_0^{\infty} \frac{\cos s(x - \pi) + \cos sx}{1 - s^2} ds \text{ when } 0 \leq x \leq \pi$$
- (b) Prove that $(a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}$
6. (a) If $u(x, y)$ satisfies $\frac{\partial^2 u}{\partial x^2} - 3 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ and $u = -x^2$ and $\frac{\partial u}{\partial y} = 0$ for $y = 0$ and all x , find the value of $u(0, 1)$
 (b) Evaluate (i) $\int_0^{\infty} \frac{x^3}{(1-x)^5} dx$ (ii) $\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}} \theta d\theta$
 (c) Show that $u(x, t) = e^{-kt} \sin x$ satisfy the heat equation $u_{xx} = \frac{1}{k} u_t$

DEPARTMENT OF MATHEMATICAL SCIENCES

Examination: B.Tech

Course Code: MTS 316

Course Title: Engineering Mathematics II

Semester: First

Session: 2016/2017

Instruction: Answer any three(3) questions

Time Allowed: 2 hours

- 1(a) Use Finite Fourier transforms to solve the IBVP

$$\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}, \quad 0 < x < \pi, t > 0$$

$$U(0, t) = U(\pi, t) = 0, \quad t > 0$$

$$U(x, 0) = 2x, \quad 0 < x < \pi.$$

and give physical interpretation of the problem.

- 1(b) Use the method of separation of variables to solve the IBVP

$$\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}, \quad 0 < x < 1, t > 0$$

$$U(0, t) = U(1, t) = 0, \quad t > 0$$

$$U(x, 0) = \sin 3\pi x - 2 \sin 5\pi x, \quad 0 < x < 1.$$

- 2(a) State three classes of partial differential equations with an example each.

- (b) Form a partial differential equation from
- $x^2 + y^2 + (z - c)^2 = a^2$
- , taking
- z
- as the dependent variable.

- (c) Solve the differential equation
- $\frac{\partial^2 u}{\partial x^2} = 6x^2(2y - 1)$
- given the boundary conditions that at
- $x = 0, \frac{\partial u}{\partial x} = \sin 2y$
- and
- $u = \cos y$
- .

- 3(a) Define Beta, Gamma and Error Functions. Prove that
- $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}, m, n > 0$
- .

- (b) Evaluate (i)
- $\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta$
- , (ii)
- $\int_0^3 \frac{x^3}{\sqrt{3-x}} dx$
- and (iii)
- $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$
- .

- (c) Show that the eigenvalues of Sturm-Liouville system are real.

- 4(a) Find the Fourier transform of the function
- $f(x) = \begin{cases} 1 & (|x| \leq a) \\ 0 & (|x| > a) \end{cases}$
- .

- (b) Find the Fourier sine and cosine integral representation of
- $f(x) = \begin{cases} 1 & 0 < x < \pi \\ 0 & x > \pi \end{cases}$
- .

- (c) let
- $f(x)$
- be a function of periodic
- 2π
- such that
- $f(x) = \begin{cases} 1 & -\pi < x < 0 \\ 0 & 0 < x < \pi \end{cases}$
- .

- (i) Show that Fourier series for
- $f(x)$
- in the interval
- $-2\pi < x < 2\pi$
- is

$$\frac{1}{2} - \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right].$$

- (ii) By using appropriate value of
- x
- , show that
- $\frac{\pi}{4} = \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right]$
- .

- 5 (a) Consider the temperature distribution $u(x, y)$ over a thin rectangular body D in which the sides $x = 0$, $x = l$, $y = 0$ are kept at zero temperature and the side $y = b$ is maintained at given temperature distribution $f(x)$ for $0 \leq x \leq l$. If the physical phenomenon is modeled as

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Subject to

$$u(0, y) = 0, \quad u(l, y) = 0 \quad 0 \leq x \leq l$$

$$u(0, b) = f(x) = \sin\left(\frac{n\pi x}{l}\right), \quad u(x, 0) = 0 \quad 0 \leq y \leq b$$

Use the method of separation of variables to obtain the temperature distribution $u(x, y)$.

- (b) Consider the Laplace equation

$$u_{xx} + u_{yy} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Subject to

$$u(x, 0) = 0 \quad 0 < y < L$$

$$u(x, L) = \begin{cases} 1 & -a < x < a \\ 0 & \text{Otherwise} \end{cases}$$

Use the method of Fourier transform to obtain the solution $u(x, y)$

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
B. TECH DEGREE EXAMINATION
SECOND SEMESTER 2012/2013
MTS316 - ENGINEERING MATHEMATICS II

Instruction: Answer any four questions.

Time allowed: 2 hrs

1a) Evaluate the following integrals using Gamma functions $\Gamma(n)$

$= \int_0^{\infty} x^{n-1} e^{-x} dx$, (for $n > 0$), where, $\Gamma(n)$ converges for $n > 0$ (positive n).

(i) $\int_0^{\infty} x^{-\frac{1}{2}} e^{-x} dx$ (ii) $\int_0^1 x^2 \left(\log \frac{1}{x} \right) dx$ (iii) $\int_0^{\infty} x^{5/2} e^{-x^2} dx$

b) Prove that $\Gamma(n) = \Gamma\left(\frac{n+1}{n}\right)$. Hence, evaluate (i) $\Gamma\left(-\frac{1}{2}\right)$, (ii) $\Gamma\left(\frac{8}{3}\right)$

c) Prove that $\int_0^{\infty} e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$ and deduce that $\sqrt{\pi} = \Gamma\left(\frac{1}{2}\right)$

2a) Express the following integrals as β - functions

(i) $\int_0^1 \frac{x^3}{(1+x)^5} dx$, (ii) $\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}} \theta d\theta$, (iii) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

b) Prove that $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$, where $B(m, n)$ is a beta and Γ gamma functions. Hence, evaluate $\int_0^1 \frac{x^4}{\sqrt{1+x^2}} dx$

3a) Find the Fourier series of period 2π which gives a function equal to 0 for $0 < x < \pi$

and equal to $\cosh x$ for $-\pi < x < 0$. By considering the values of the series at the points of discontinuity, deduce that

$$\sum_{r=1}^{\infty} \frac{(-1)^r}{1+r^2} = \frac{1}{2} \left(\frac{2\pi}{\sinh \pi} - 1 \right) \text{ and } \sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{1}{2} \left(\frac{2\pi}{\tanh \pi} - 1 \right)$$

b) Given that

$$f(x) = \begin{cases} -1 & \text{for } -\pi < x < 0, \\ 0 & \text{for } x = 0, \\ +1 & \text{for } 0 < x < \pi, \end{cases}$$

$-\pi < x < \pi$. Show that $f(x) = \frac{4}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$

4a) Show that the Fourier expansion of

$$f(x) = \begin{cases} 1 + (\alpha/\pi) & \text{for } -\pi \leq x \leq 0, \\ 1 - (\alpha/\pi) & \text{for } 0 \leq x \leq \pi \end{cases}$$

in the range $-\pi \leq x \leq \pi$ is

$$f(x) = \frac{1}{2} + \frac{4}{\pi^2} \left(\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right)$$

Hence deduce that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}$$

b) Show that $|\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{\text{even } r} \frac{\cos rx}{r^2 - 1}$

5a) (i) Give two differences between ordinary differential equation (ODE) and partial differential equation (PDE). (ii) Give two examples each of ODE and PDE.

b) When is a PDE said to be (i) Elliptic, (ii) Parabolic and (iii) Hyperbolic. Give an example in each case.

c) Verify that $U(x,t) = \sin x \cos kt$ is a solution of the wave equation

$$U_{xx} = \frac{1}{k^2} U_{tt}$$

6a) Give two examples each of (i) Linear (ii) Non-linear and (iii) Quasi-linear PDE.

b) Solve the equation; $\frac{\partial^2 U}{\partial x \partial y} = \sin(x+y)$ given that at $y=0$, $U_x = 1$ and at $x=0$, $U = (y-1)$

$$-\frac{1}{u} \left[\frac{-\cos(r+1)u}{r+1} + \frac{\cos(r-1)u}{r-1} \right] + \frac{1}{r+1} - \frac{1}{r-1}$$

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
B. TECH DEGREE EXAMINATION
SECOND SEMESTER 2013/2014
MTS316 - ENGINEERING MATHEMATICS II

INSTRUCTION: Answer Four Questions. At least ONE from each Section; Time allowed: 2hrs.

SECTION A

1a. Gamma functions $\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$, (for $n > 0$), where, $\Gamma(n)$ converges for $n > 0$ (positive n).

Show that, $\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$. [Hint: use $x = x^2$]

1b. Hence, evaluate (i) $\Gamma\left(\frac{7}{2}\right)$, (ii) $\Gamma\left(-\frac{3}{2}\right)$,

1c. Evaluate the following integrals using Gamma-functions $\Gamma(n)$

(i) $\int_0^{\infty} x^3 e^{-4x} dx$, (ii) $\int_0^1 x^2 \left(\log \frac{1}{x}\right) dx$

2a. Prove that, $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$, where $B(m, n)$ is a beta and Γ gamma functions.

2b. Hence, evaluate, (i) $\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta$, (ii) $\int_0^3 \frac{x^3}{\sqrt{3-x}} dx$ and (iii) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

SECTION B

3a. Define a partial differential equation and give two examples.

3b. Define the following classes of PDE, (i) elliptic, (ii) parabolic and (iii) hyperbolic.

3c. Solve the partial differential equation;

$$\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} + 3 \frac{\partial z}{\partial x} + 2z = 0$$

4a. Let z be a function of two variables x and y such that $f(x, y, z, a, b) = 0$, where a and b are two arbitrary constants. Suppose that two arbitrary functions U and V are connected by the relation $F(U, V) = 0$, where $U = U(x, y, z)$ and $V = V(x, y, z)$. Show that $Pp + Qq = R$, where;

$$P = \frac{\partial U}{\partial z} \times \frac{\partial V}{\partial y} - \frac{\partial U}{\partial y} \times \frac{\partial V}{\partial z}, Q = \frac{\partial U}{\partial x} \times \frac{\partial V}{\partial z} - \frac{\partial U}{\partial z} \times \frac{\partial V}{\partial x}, R = \frac{\partial U}{\partial y} \times \frac{\partial V}{\partial x} - \frac{\partial U}{\partial x} \times \frac{\partial V}{\partial y}, p = \frac{\partial z}{\partial x} \text{ and } q = \frac{\partial z}{\partial y}$$

4b. Solve the equation $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 0$, $0 \leq x \leq \pi$, subject to the following boundary conditions;

(i) $U(0, y) = 0$ (ii) $U(\pi, y) = 0$ (iii) $U(x, 0) = 1$ (iv) $U(x, y) \rightarrow 0$ as $y \rightarrow \infty$

SECTION C

5a. Find the Fourier series of period 2π which gives a function equal to 0 for $0 < x < \pi$ and equal to $\cosh x$ for $-\pi < x < 0$. By considering the values of the series at the points of discontinuity, deduce that

$$\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{1+r^2} = \frac{1}{2} \left(\frac{2\pi}{\sinh x} - 1 \right) \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{1}{2} \left(\frac{2\pi}{\tanh x} - 1 \right)$$

5b. Given that, for $-\pi < x < \pi$,

$$f(x) = \begin{cases} -1 & \text{for } -\pi < x < 0, \\ 0 & \text{for } x = 0, \\ +1 & \text{for } 0 < x < \pi. \end{cases}$$
 Show that $f(x) = \frac{4}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$

6a. Show that the Fourier expansion of

$$f(x) = \begin{cases} 1 + (x/\pi) & \text{for } -\pi \leq x \leq 0, \\ 1 - (x/\pi) & \text{for } 0 \leq x \leq \pi \end{cases}$$

 in the range $-\pi \leq x \leq \pi$ is $f(x) = \frac{1}{2} + \frac{4}{\pi^2} \left(\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right)$. Hence deduce that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}$$

6b. Show that

$$|\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{r=1}^{\infty} \frac{\cos rx}{r^2 - 1}$$

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DEPARTMENT OF MATHEMATICAL SCIENCES
B.TECH DEGREE IN MATHEMATICS
SECOND SEMESTER, 2008/2009 SESSION
MTS316 - ENGINEERING MATHEMATICS.

NOVEMBER 2009.

INSTRUCTIONS: ANSWER ANY FOUR (4) QUESTIONS.

TIME : 2HOURS

1. If given $\Gamma(n) = \int_0^\infty t^{n-1} e^{-t} dt$, show that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.
Also show that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta,$$

then solve

(i). $\int_0^2 x(8-x^3)^{1/3} dx$, (ii). $\int_0^{\pi/2} \sin^6 \theta d\theta$, (iii). $\int_0^{\pi/4} \sin^3 2\theta \cos^6 2\theta d\theta$.

- 2.a. Given a Sturm-Liouville's problem

$$(e^{2x} y')' + e^{2x} (\lambda + 1) y = 0,$$

with conditions

$$y(0) = y(\pi) = 0.$$

- (i). Find the eigenvalues and the corresponding eigenfunctions of the problem.
(ii). Determine if the eigenfunctions are orthogonal in $I: 0 \leq x \leq \pi$.

2. b. Given the function $f(\theta) = \frac{e^{im\theta}}{\sqrt{2\pi}}$, $m = 0, \pm 1, \pm 2, \pm 3, \dots$ in $(-\pi, \pi)$. Show that $f(\theta)$ is orthogonal, then state the orthonormal set.

Hint: use the relation

$$\begin{aligned} e^{im\theta} &= \cos m\theta + i \sin m\theta \\ e^{-im\theta} &= \cos m\theta - i \sin m\theta \end{aligned}$$

3. a(i). If $\mathcal{L}[f(t)] = \int_0^\infty e^{-st} f(t) dt = F(s)$,

show that $F(s) = \frac{n!}{s^{n+1}}$ whenever $f(t) = t^n$.

(ii). Show that $\mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$.

3. b. In the circuit, Figure 1, if at time $t = 0$, the key is closed. Assuming that the charge on the capacitor is zero at $t = 0$. Find the charge and current at any later time. Assume R, C, E are constants.

4. a(i). Solve the pair of simultaneous equations for x and y

$$\frac{dy}{dt} - x = e^t$$

$$\frac{dy}{dt} + y = e^{-t}$$

given that $x(0) = y(0) = 0$.



Figure 1: Electric circuit

(ii). If $F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$, n is a positive integer. Find the coefficients a_0 , a_n and b_n .

4. b. Solve the equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 \leq x \leq \pi$$

subject to (i). $u(0, y) = 0$, (ii). $u(\pi, y) = 0$, (iii). $u(x, 0) = 1$, and

(iv). $u(x, y) \rightarrow 0$, as $y \rightarrow \infty$.

5. a. Use the method of separation of variables to solve the equation

$$\frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t},$$

given that $v = 0$ when $t \rightarrow \infty$, as well as $v = 0$ at $x = 0$ and $x = l$.

b. Solve $\tan x \frac{dy}{dx} + y = \sec x$.

6. a. Consider the second order linear differential equation $(a_0 D^2 + a_1 D + a_2)y = 0$. Show that the Abel's identity of this equation is given by

$$W = C e^{-\int a_1/a_0 dx}.$$

Hint: let y_1 and y_2 be the solution of the equation above.

b. Obtain the general solution of $(D^2 + 4D + 4)y = 6 \sin 2x$, by the method of determined coefficient. ♣♦♥♠¹

¹Examiners:

Dr. F.B. AGUSTO

Mr. AWODOLA

Mr. OWOLABI

FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF MATHEMATICAL SCIENCES
TEST

Course Code: MTS 316

Session: 2016/2017

Instruction: Answer any all questions

Time Allowed: 45mins

- 1(a) Use the separation of variables to solve the heat flow problem

$$\frac{\partial U}{\partial t} = 7 \frac{\partial^2 U}{\partial x^2}, \quad 0 < x < \pi, \quad t > 0$$

$$U(0, t) = U(\pi, t) = 0, \quad t > 0$$

$$U(x, 0) = 3\sin 2x - 6\sin 5x, \quad 0 < x < \pi$$

- 2(a) Show that the Fourier series for $f(x) = x$ in the interval $-\pi < x < \pi$ is

$$2(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \dots).$$

- (b) By using an appropriate value to x , show that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$.

- 3(a) Prove that $\Gamma(n+1) = n\Gamma(n) = n!$ for $n > 0$.

- (b) Show that the eigenvalues of Sturm-Liouville system are real.

- 4(a) Form a partial differential equation from $z = f(x^2 - y^2)$ taking z as the dependent variable.

- (b) Verify that $u = e^{-y} \cos x$ is a solution of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
DEPARTMENT OF MATHEMATICAL SCIENCES

NAME OF EXAMINATION: B. Tech Industrial Mathematics

SEMESTER / SESSION: Second Semester, 2009/2010

COURSE: MTS316 – Engineering Mathematics II

INSTRUCTION: Answer any **four** questions

TIME ALLOWED: 2 hours

1a. (i) Prove that $\Gamma(n+1) = n\Gamma(n) = n!$, $n > 0$

(ii) Evaluate $\int \frac{dx}{\sqrt{-\ln x}}$

b. Show that the eigenvalues of a Sturm-Liouville system are real.

2a. Prove that $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$, $m, n > 0$.

b. A particle is attracted toward a fixed point O with a force inversely proportional to its instantaneous distance from O. If the particle is released from rest, find the time for it to reach O. (Note that at time $t = 0$ the particle is located on the x-axis at $x = a > 0$ and O is the origin.)

3a. Find the Fourier series for the function

$$f(x) = \begin{cases} 0, & -\pi < x < -\frac{\pi}{2} \\ 4, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < x < \pi \end{cases} ;$$

$$f(x+2\pi) = f(x)$$

b. Determine the Fourier series to represent the periodic function

$$f(x) = \begin{cases} -x, & -\pi < x < 0 \\ 0, & 0 < x < \pi \end{cases} ;$$

$$f(x+2\pi) = f(x)$$

4a. Solve $\frac{\partial^2 z}{\partial y^2} - 16z = 0$.

b. Obtain a solution of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the boundary conditions;

$$u = \sin y \quad \text{when} \quad x = 0 \quad \text{for all } y$$

$$u \rightarrow 0 \quad \text{when} \quad x \rightarrow \infty.$$

5a. Solve the boundary value problem

$$\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial x} = 2x, \quad x > 0, \quad y > 0$$

$$u(x, 0) = x^2 + \cos x, \quad x > 0,$$

$$\frac{\partial u(0, y)}{\partial y} = 2y, \quad y > 0.$$

b. Find, using separation of variables, all the solutions of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$

6a. Solve the Laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for the interior of a rectangle with the boundary conditions $u(0, y) = 50$, $u(a, y) = 0$, $u(x, 0) = 0$, $u(x, b) = 0$

b. Find the solution of the heat equation $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$, $k > 0$ for the initial and boundary conditions $u(x, 0) = x$, $u(0, t) = 0$, $u(L, t) = 0$

FEDERAL UNIVERSITY OF TECHNOLOGY AKURE.
DEPARTMENT OF MATHEMATICAL SCIENCES
B.TECH DEGREE IN MATHEMATICS
SECOND SEMESTER, 2005/2006 SESSION
MTS316 - ENGINEERING MATHEMATICS.

DATE _____

JULY 2006.

INSTRUCTIONS: ANSWER ANY THREE (3) QUESTIONS.

TIME : 2HOURS

1. a i. If

$$\mathcal{L}[F(t)] = \int_0^{\infty} e^{-st} F(t) dt. \text{ Show that } \mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

ii. If $s > 2$, find the Laplace inverse of

$$\frac{3}{s} - \frac{5}{s-2} + \frac{4}{s^2+4} - \frac{7s}{s^2+9}$$

b i. If

$$\mathcal{L}[F(t)] = \int_0^{\infty} e^{-st} F(t) dt \text{ Show that } \mathcal{L}[F''(t)] = s^2 \mathcal{L}[F(t)] - sF(0) - F'(0)$$

where $F''(t) = \frac{d^2 F(t)}{dt^2}$. Hence;

ii. Solve, using Laplace transform, the differential equation

$$\frac{d^2 x}{dt^2} - 2 \frac{dx}{dt} + x = e^t$$

2. a i. If

$$F(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \{A_n \cos nx + B_n \sin nx\}, \quad n > 0.$$

Find the Fourier coefficients A_0 , A_n and B_n .

ii. Find the Fourier series of $f(x)$, where

$$f(x) = \begin{cases} 0, & -\pi < x < -\frac{\pi}{2} \\ 4, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < x < \pi \end{cases}$$

b i. Prove that if the wronskian corresponding to two solutions y_1 and y_2 is zero at some point of the interval, then it is identically zero in the interval and the solutions y_1 and y_2 are linearly dependent.

ii. Show that the functions x^2 and x^3 are linearly independent solutions of the equation $x^2 y'' - 4xy' + 6y = 0$.

3. a i. Find the solution of the homogeneous differential equation

$$y'' - 3y' + 2y = 0$$

hence;

- ii. Find the general solution of the following differential equation by the method of variation of parameter:

$$y'' - 3y' + 2y = xe^x + 2x$$

- b. Find the general solution of the system of equation

$$\begin{aligned} y_1'' + y_1' - y_1 + y_2'' - 3y_2' + 2y_2 &= 0 \\ y_1' + 2y_1 + 2y_2' - 4y_2 &= 0 \end{aligned}$$

4. a. Determine if the limit of the following exist or does not exist.

i

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2 \sin y}{2x^2 + y^2}$$

ii. $\lim_{(x,y) \rightarrow (0,0)} \frac{x(\cos y - 1)}{x^3 + y^3}$

- b. Consider

$$U_t = U_{xx}$$

$$U(0, t) = U(L, t) = 0$$

$$U(x, 0) = f(x)$$

Show that

$$U(x, t) = \sum_{n=1}^{\infty} B_n e^{-\frac{n^2 \pi^2}{L^2} t} \sin \frac{n\pi}{L} x$$

where

$$B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x dx$$

Hence solve the equation

i.

$$U_t = U_{xx}$$

$$U(0, t) = U(\pi, t) = 0$$

$$U(x, 0) = \sin x$$

ii.

$$U_t = U_{xx}$$

$$U(0, t) = U(\pi, t) = 0$$

$$U(x, 0) = x(\pi - x)$$

5. i. A manufacturing company produces two models of a television set, x units of model A and y units of model B per week, at a cost of $C(x, y) = 6x^2 + 12y^2$ if it is necessary (because of shipping consideration) that $x + y = 90$ how many of each type of set should be manufactured per week to minimize cost? What is the minimum cost?
- ii. Maximize

$$f(x, y) = e^{(x^2 + y^2)}$$

Subject to

$$g(x, y) = x^2 + y - 1$$

by method of Lagrange multiplier.

- iii. Show that $(0, 0)$ is a critical point for the function $f(x, y) = x^4 y + x^2 y^4 + 1$ but that the second derivative test for the local extrema fails.

$$F(x, y) = x^4 y + \lambda(g(x, y))$$